

Cyclistic Bike Analysis

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Last Updated: June 24 2026

Objective

Identify behavioral differences between **casual** and **member** riders to support **membership-conversion strategies**.

Data Sources

- Provided by Motivate International Inc.
- 12 months of Cyclistic trip data with timestamps, ride length, rider type, and station details.
- Processed in Python (Pandas) and visualized using spreadsheets.

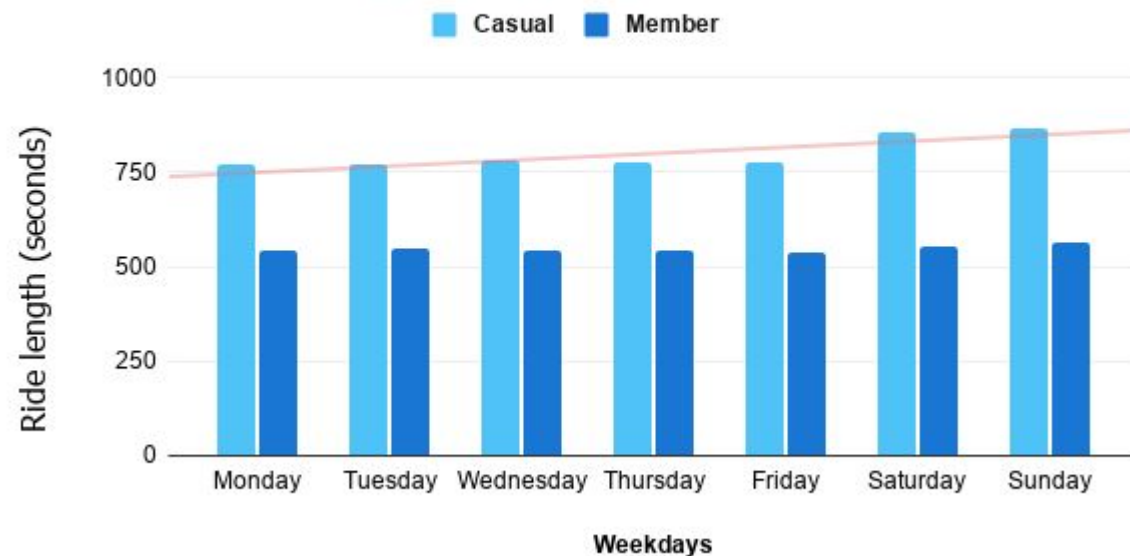
Data Cleaning

- Removed duplicates and null values
- Excluded negative ride durations
- Converted timestamps to proper date-time format
- Added new columns for ride length and weekday
- Verified data consistency before visualization

Trends

Average Ride Length by Day of Week

Casual riders tend to take longer trips, especially on weekends

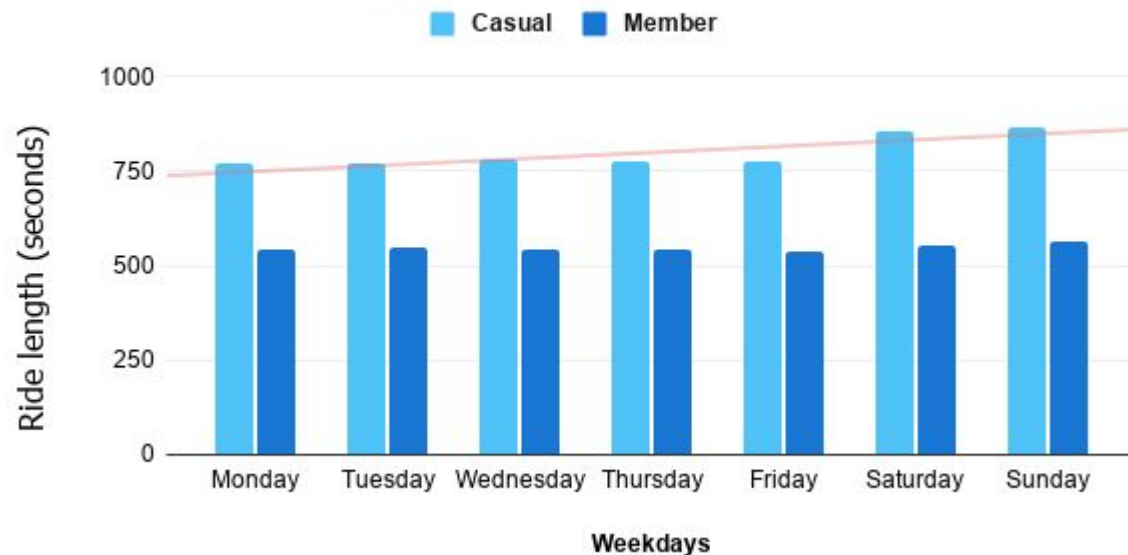


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- **Casual** riders show higher variability across weekdays

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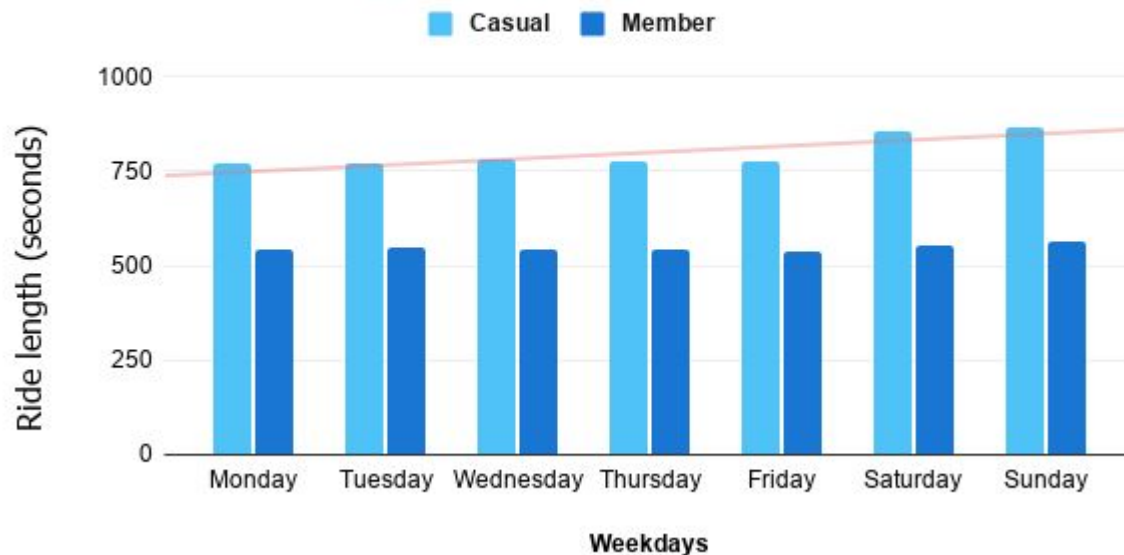


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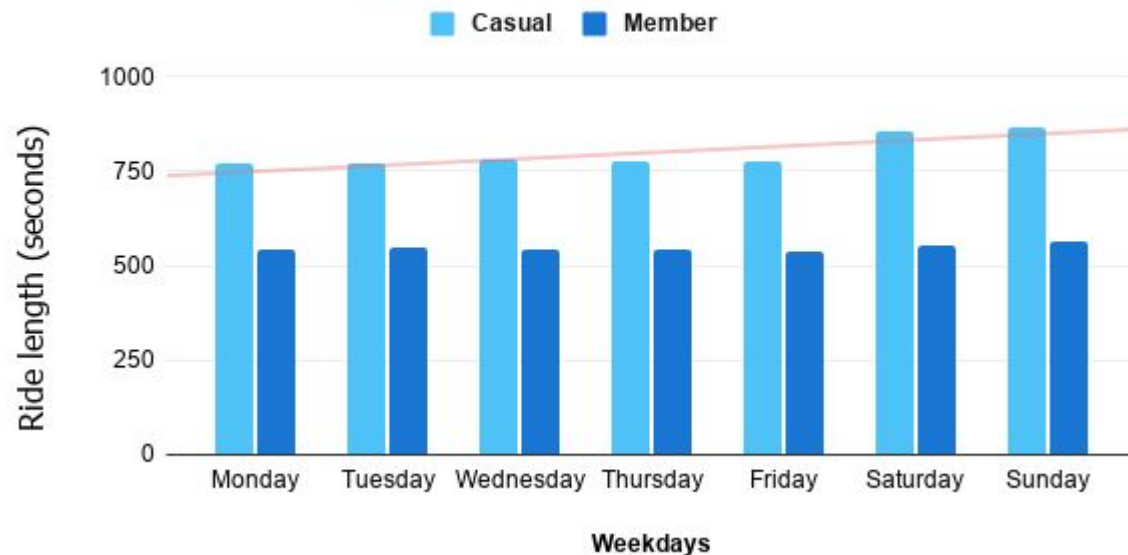


Trends

- **Casual** riders show higher variability across weekdays
- **Members** ride longer on average than **casual** riders
- **Members** maintain consistent ride duration throughout the week

Average Ride Length by Day of Week

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Conclusions

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1

Ride length reveals
clear behavioral
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2

These patterns highlight opportunities to **increase membership**, **optimize weekend operations**, and **improve overall rider experience**.

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Ride length reveals clear behavioral **differences** between **Casual** and **Member** riders.

2

These patterns highlight opportunities to **increase membership**, **optimize weekend operations**, and **improve overall rider experience**.

3

By acting on these insights, the business can drive **higher engagement**, **better resource allocation**, and stronger **long-term growth**.

Business Recommendations

- Encourage casual riders to become members by offering weekend-focused membership perks.
- Design marketing campaigns around longer weekend rides.
- Optimize bike availability for weekends when casual demand peaks.
- Adjust pricing or promotions to balance weekday usage.

Thank you